Project Report

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Subject Name: IOT Lab **Subject Code:** 24CAP-751

Aim:

IoT Based Smart Trolley System for automatic billing using RFID technology

Objectives:

1. To develop a system for automatic item billing using RFID.

2. To reduce long queues at billing counters.

3. To provide real-time total amount display to the user.

4. To implement budget monitoring and item removal functionality.

Sno	Name of Component	Quantity
1	Arduino UNO	1
2	MFRC522 RFID Module	1
3	RFID Tags/Cards	2–3
4	OLED Display (SSD1306)	1
5	Push Buttons	3
6	LEDs	3
7	Resistors (220 Ohm)	3

8	Buzzer	1
9	Breadboard	1
10	Jumper Wires	As required

Details of Components:

1. Arduino UNO:

The Arduino UNO is the main controller of the system. It processes the input from the RFID module and buttons, and controls the output devices such as OLED display, LEDs, and buzzer. It acts as the brain of the smart trolley system.

2. RFID Module (MFRC522):

The RFID module is used to scan RFID tags attached to products. Each tag has a unique ID which is read by the module and sent to the Arduino for processing

3. OLED Display (SSD1306):

The OLED display is used to show real-time information such as item added, item removed, total amount, and final bill.

4. Push Buttons:

Three buttons are used for user interaction:

* Budget button → to check total amount
* Remove button → to remove item
* Final/Clear button → to display final bill or reset

5. LEDs:

LEDs provide visual indication:

* LED 1 → Item added

* LED 2 → Item removed

* LED 3 → Final bill

6. Buzzer:

The buzzer gives audio alerts when an item is scanned or when the budget exceeds.

7. Breadboard & Jumper Wires:

Used to connect all components without soldering and make the circuit flexible**.**

Block Diagram of Designed Model:

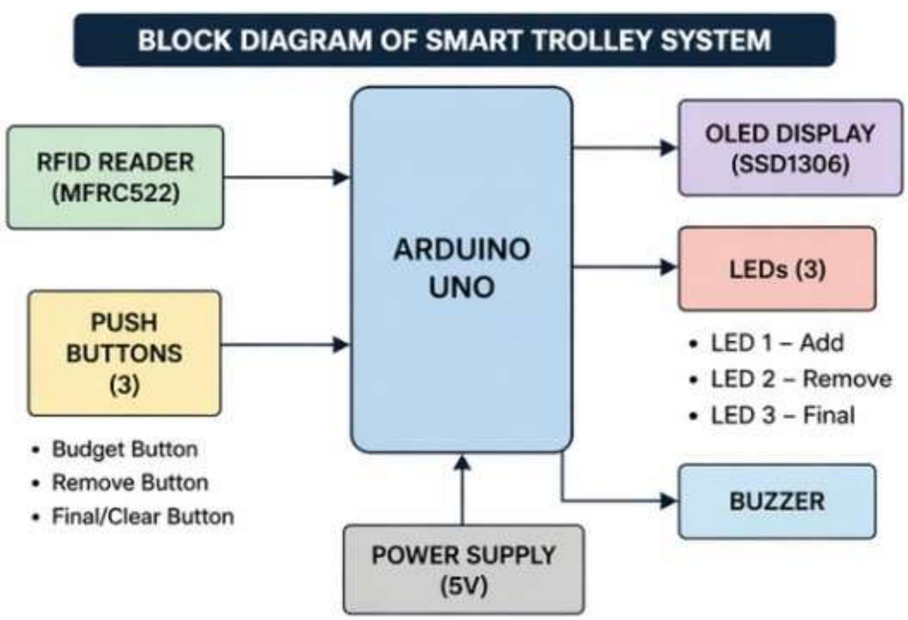


Figure 1: Block Diagram of Smart Trolley System

This diagram shows the overall working of the system where RFID module sends data to Arduino, which processes it and displays output on OLED. Buttons, LEDs, and buzzer are connected for user interaction and indication.

Explanation of Block Diagram:

* RFID module scans the product tag

* Arduino receives UID and processes data

* Total amount is updated

* OLED display shows the result

* Buttons allow user interaction

* LEDs and buzzer provide indication

Working of Designed Model:

* When a user scans an RFID tag, the system reads the unique ID and adds the item price to the total amount.

* The OLED display shows the updated bill instantly.

* The remove button subtracts the item price from the total.

* The budget button displays the current total and alerts if the limit is exceeded.

* The final button shows the final bill and can reset the system.

* LEDs and buzzer provide visual and audio feedback for each action.

Pictures of Prototype:

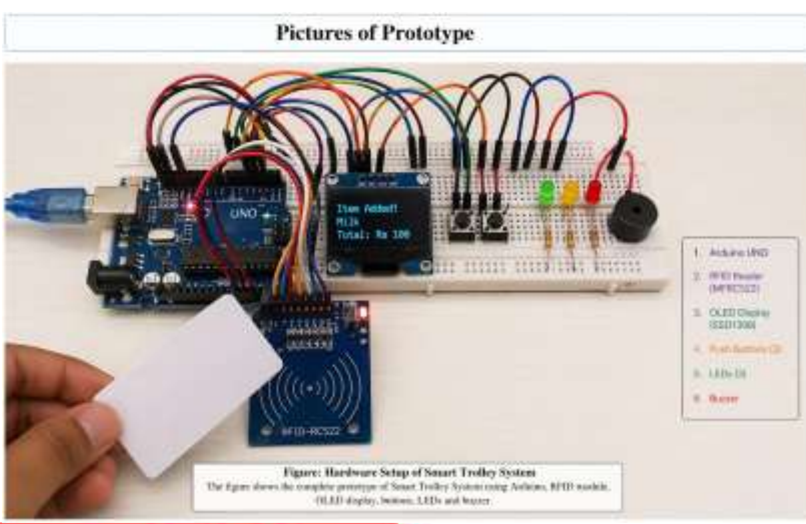
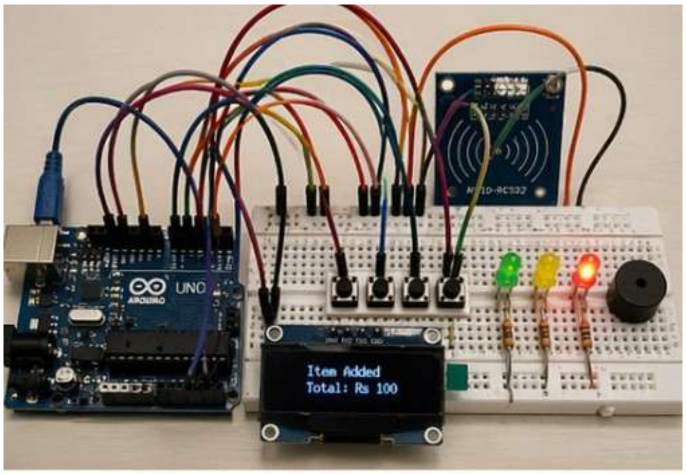


Figure: Hardware Setup of Smart Trolley System
The figure shows the complete prototype of Smart Trolley System using Arduino, RFID module, OLED display, buttons, LEDs and buzzer.

Figure : Designed Model



Output of Designed Model/Prototype

* OLED shows item status and total amount

* LEDs indicate actions

* Buzzer gives alert

Figure : Output Display

Figure 4: OLED Display Output

Learning Outcomes (What I have learnt):

1. Understanding of RFID technology and its applications
2. Hands-on experience with Arduino and IoT components
3. Implementation of real-time embedded systems
4. Circuit design and hardware integration skills

Figure 4 : Output on BLYNK Cloud